

Ankit Barana

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EDUCATION

Oberlin College, Oberlin, OH

Computer Science and Neuroscience Double Major

Expected: May 2025

Major GPA: 3.95/4.0

PROJECTS

Leslie Kwakye Lab Website - A website built with Jekyll for the Sensory Neuroscience Lab at Oberlin College

Shortify - A URL-shortener built in Rust using Yew and Rocket shortened and customized URLs

RESEARCH EXPERIENCE

Research Assistant

January 2024

Mentor: Albert Powers, MD-PhD (Powers Lab, Yale School of Medicine, New Haven, CT)

- Developed a React/Node.js web application with OpenAI API to visualize patient-reported hallucinations
- Collaborated with other lab members in developing and debugging multiple React-based experiments
- Conducted extensive literature review to identify relevant studies and inform our experimental design and development
- Documented progress on weekly basis

Research Assistant

June 2022 - Present

Mentor: Leslie Kwakye, PhD (Sensory Neuroscience Lab, Department of Neuroscience, Oberlin, OH)

- Assisted in the development of methodology in the study exploring interplay between attention and multisensory integration of complex sensory stimuli, using psychophysics, eye-tracking, and electroencephalography (EEG)
- Develop the experimental stimuli using Adobe Premiere and Adobe Media Encoder, code experiment using the Experiment Builder software, and made 20,000 audio files with Matlab, integrating Eye-tracking software with EEG, and using the eye-tracking analyzing software
- Analyze EEG and Eye-tracking data using BrainVision Encoder and EyeLink DataViewer, determine appropriate eye-tracking variables (fixation time, saccades, eye-blinks, etc.) to measure, and develop analysis templates for filtering and analyzing psychophysical data obtained from multiple experimental blocks
- Conduct extensive literature review of the findings and critique the limitations of the papers at weekly meetings

WORK EXPERIENCE

Resident Assistant, Office of Residence Life, Oberlin College, OH

August 2023 - Present

- Plan and implement weekly community-building strategies ensuring a safe and inclusive learning environment
- Actively resolve resident conflicts to ensure a conflict-free dorm environment with zero room change requests

Student Assistant, Office of the President, Oberlin College, OH

August 2022 - December 2022

- Managed email correspondence, phone inquiries, and office supplies for the President's Office
- Assisted in planning and executing important events like intercollegiate dinners and high-level guest meetings

Neuroimaging Internship

June 2021 - July 2021

Mentor: Dr. Vinod Sharma, MD (Associate Professor of Neurosurgery at SMS Medical College, Jaipur, India)

- Interacted with over 20 patients daily and gained experience in Magnetic Resonance Imaging (MRI) and Computerized Tomography (CT) scan analysis
- Explored the diagnosis, treatment, and prevention of neurosurgical conditions like PIVD, spinal dysraphism, foraminal stenosis, dermoid, and neurocysticercosis, among others

LEADERSHIP

Organizer/Conductor, Smartphone Workshops

July 2021 - August 2021

- Digitally empowered the rural communities in Alwar with smartphone skills through tailored workshops
- Fostered financial inclusion by focussing on UPI payment systems and e-commerce strategies

- Collaborated with the local panchayat (village council) to coordinate logistics and maximize outreach

Founder, Neuronauts - An International Neuroscience Club

October 2019 - May 2021

- Oversaw a global virtual group of over 50 neurotech aspirants and professionals in the field from across 20 countries
- Facilitated collaborations between students and professionals via workshops, mentorships, and guidance

Event Organizer, Next Genius Foundation, Mumbai, India

August 2021

- Organized the Awards and Interviews Ceremony for the Next Genius Foundation, an organization funding talented scholarship applicants to pursue higher education
- Designed an interactive experience for 240+ individuals interviewees and monitored their progress throughout the process
- Oversaw event planning and execution, ensured smooth functioning of interactive presentations and videos, and guided participants throughout the awards and interviews ceremony

Social Service Assistant, Umang School, Jaipur, India

Jan 2021

- Organized educational, and recreational activities at Umang school, an organization aimed at developing self-reliance in the children with cerebral palsy, autism, and mental disability
- Encouraged group activities focused on developing an atmosphere of communal belonging and acceptance

CONFERENCES AND SYMPOSIA PRESENTATIONS

Oral Presentation at Oberlin Summer Undergraduate Research Fellowship (OSURF) - Featured Researcher Aug 2022, 2023
Oberlin College, OH

- Titled “How does distraction influence multisensory speech integration: A study on neural mechanisms ”
- With Arohi Dandawate, Sarah Mia Liberatore, Neuroscience Department, Oberlin College.

Poster Presentation at the Midwest/Great Lakes Undergraduate Research Symposium in Neuroscience

Oct 2022

Baldwin Wallace University, Berea, OH

- Titled “Neural Mechanisms of Attentional Alteration on Multisensory Speech Perception: Eye-Tracking, Psychophysical, and Steady-State Analysis”
- With Arohi Dandawate, Neuroscience Department, Oberlin College.

SKILLS

Research: MS Office, Latex, Adobe-Suite, Canva, SPSS, GraphPad Prism, Experiment Builder, Eyelink Plus, BrainVision, Eprime-3

Programming: Java, R, Python Rust, Racket, Matlab, HTML, CSS, Tailwind, and Typescript

Frameworks: React, Next, Rocket.rs, Yew, Jekyll, Tailwind CSS, Rusqlite

Tools: Git & Github, Command Line, Postman, Figma, Lucidchart

Languages: Hindi (Native proficiency), English (Native proficiency)